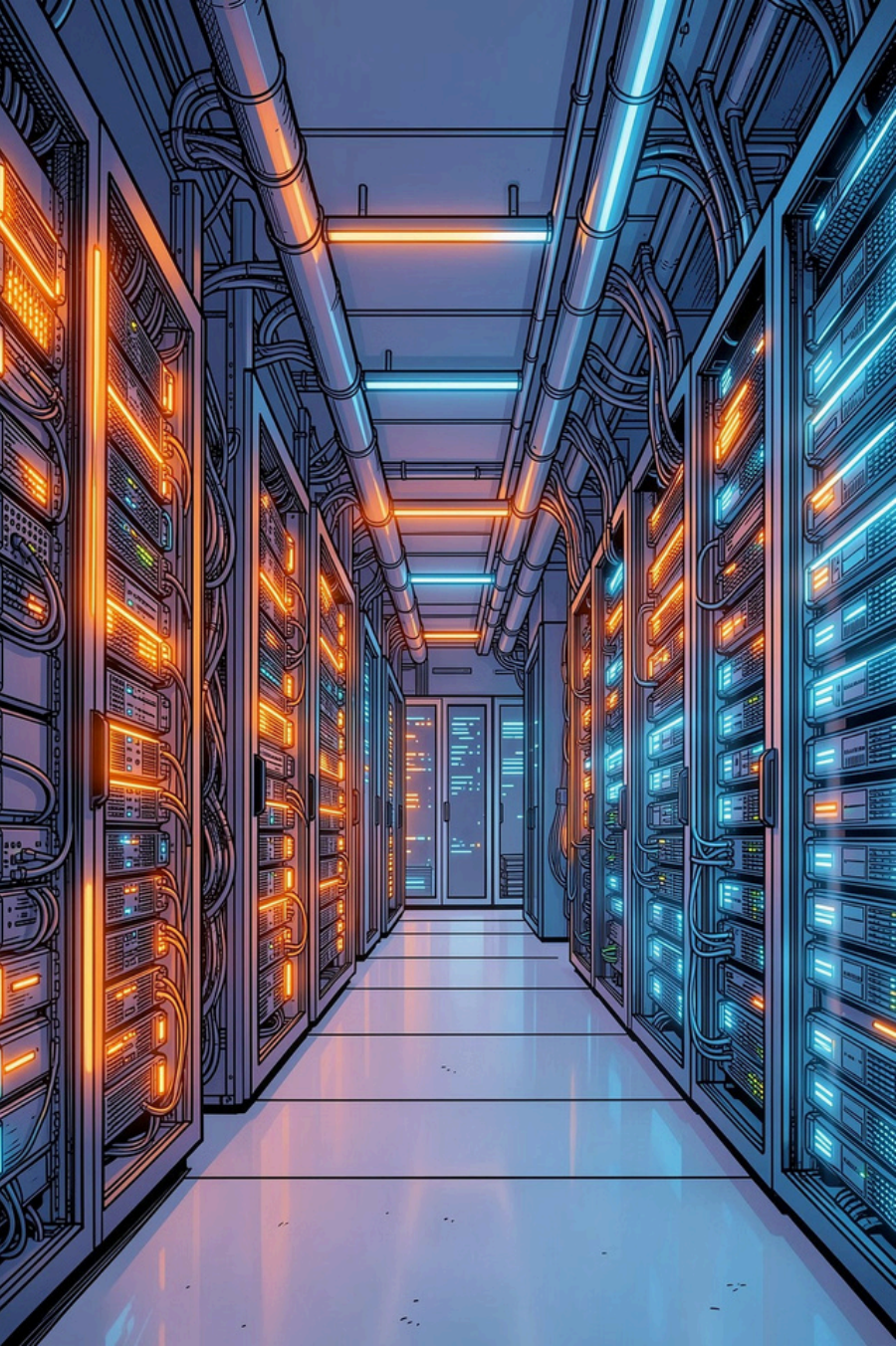




Hosting a Static Portfolio Website on AWS Cloud

Using Amazon S3 with Monitoring and Logging Services

AWS PROJECT

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Introduction

A static portfolio website is a developer's digital identity. Hosting it on **Amazon S3** eliminates server management, reduces costs, and delivers high availability — all while integrating AWS-native monitoring and security services.

What We Built

A publicly accessible portfolio site hosted on S3 with real-time monitoring and audit logging.

Why AWS S3

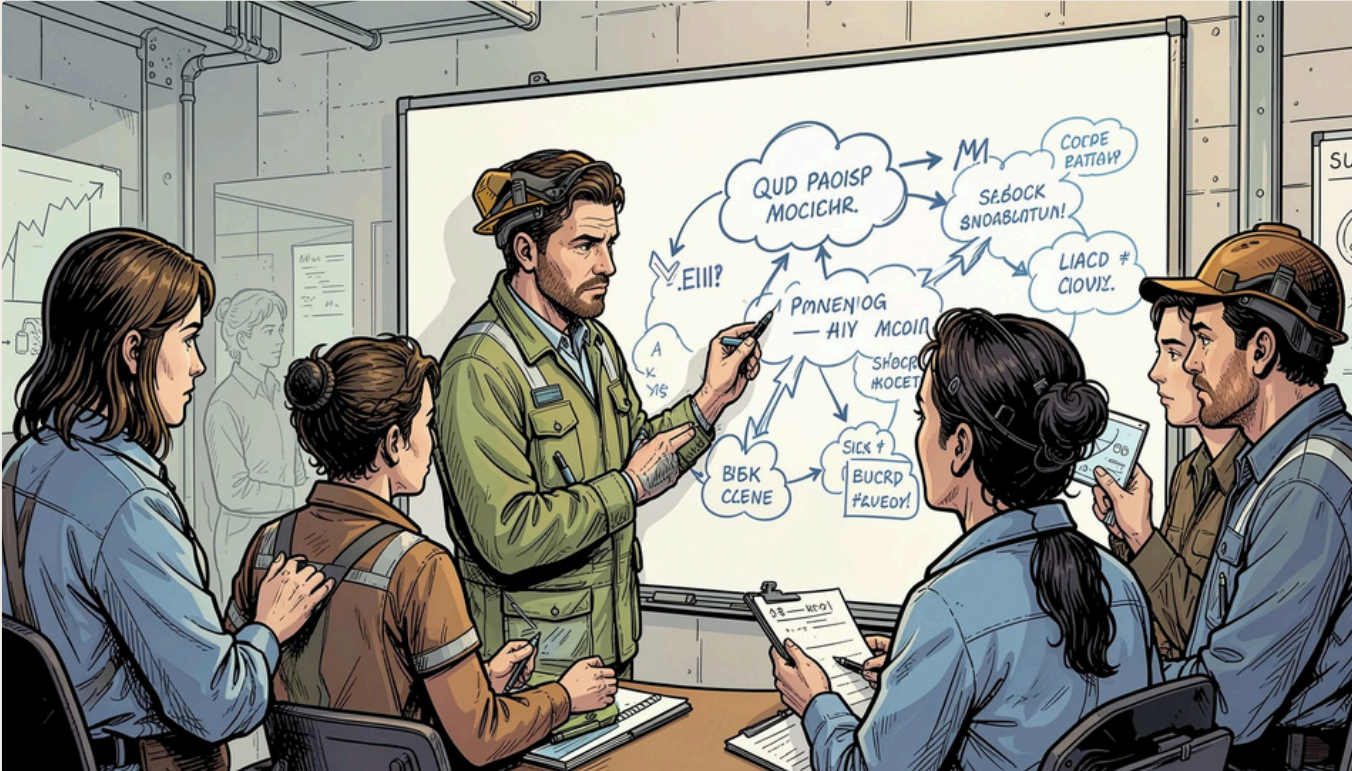
99.999999999% durability, serverless architecture, and pay-as-you-go pricing make it ideal for static hosting.

Integrated Services

CloudWatch, CloudTrail, SNS, and Systems Manager work together for observability, security, and future scalability.



Problem Statement & Objectives



The Challenge

Traditional hosting requires server

provisioning, ongoing maintenance, and significant cost. Students and developers need a **reliable, low-cost, and scalable** solution to showcase their work.

Project Objectives

- Host a static portfolio website using **Amazon S3**
- Implement **CloudWatch** for real-time monitoring
- Enable **CloudTrail** for activity audit logging
- Configure **SNS** for email-based alerts
- Plan for **Systems Manager** for future infrastructure management

AWS Services Used



Amazon S3

Object storage for hosting HTML, CSS, and JS files with built-in static website hosting.



CloudWatch

Monitors bucket metrics, request counts, and errors in real time.



CloudTrail

Records all API calls and user activity for security auditing and compliance.



SNS

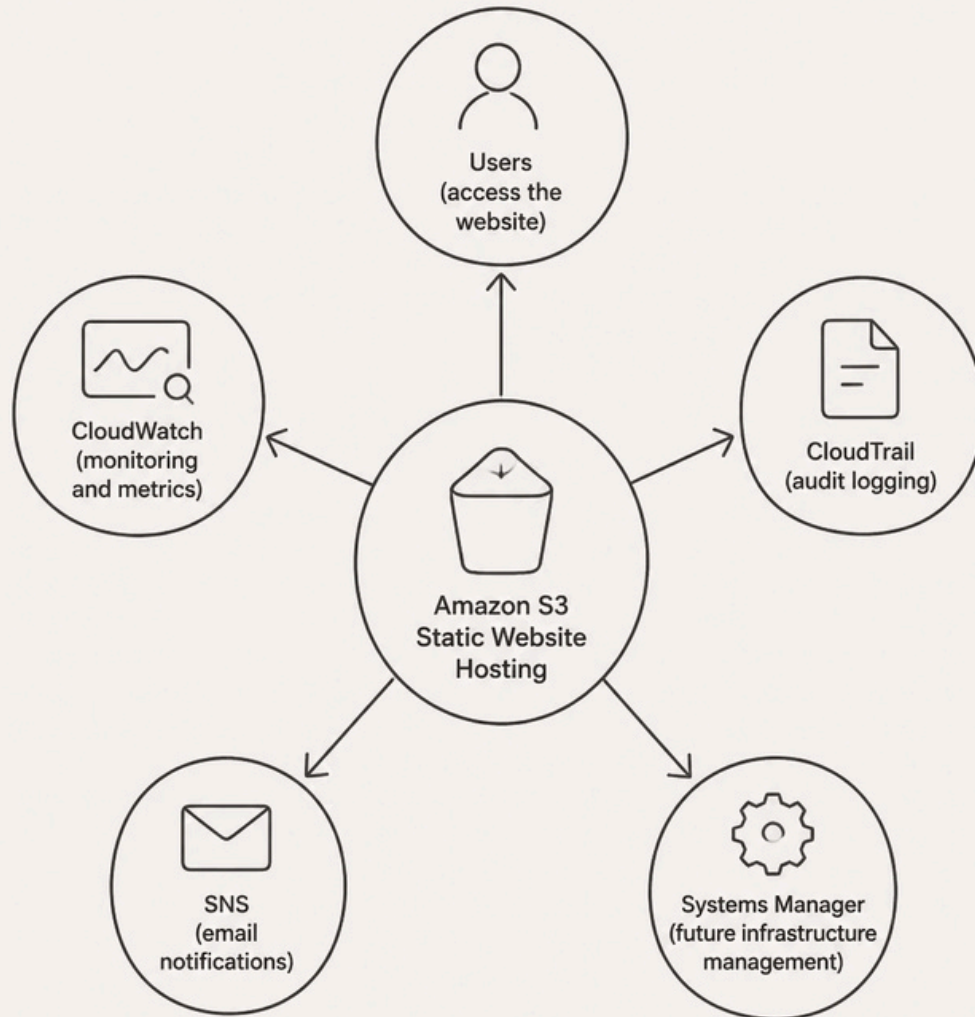
Sends message alerts for critical events like unauthorized access or threshold breaches.



Systems Manager

Enables centralized infrastructure management and automation for future expansion.

System Architecture

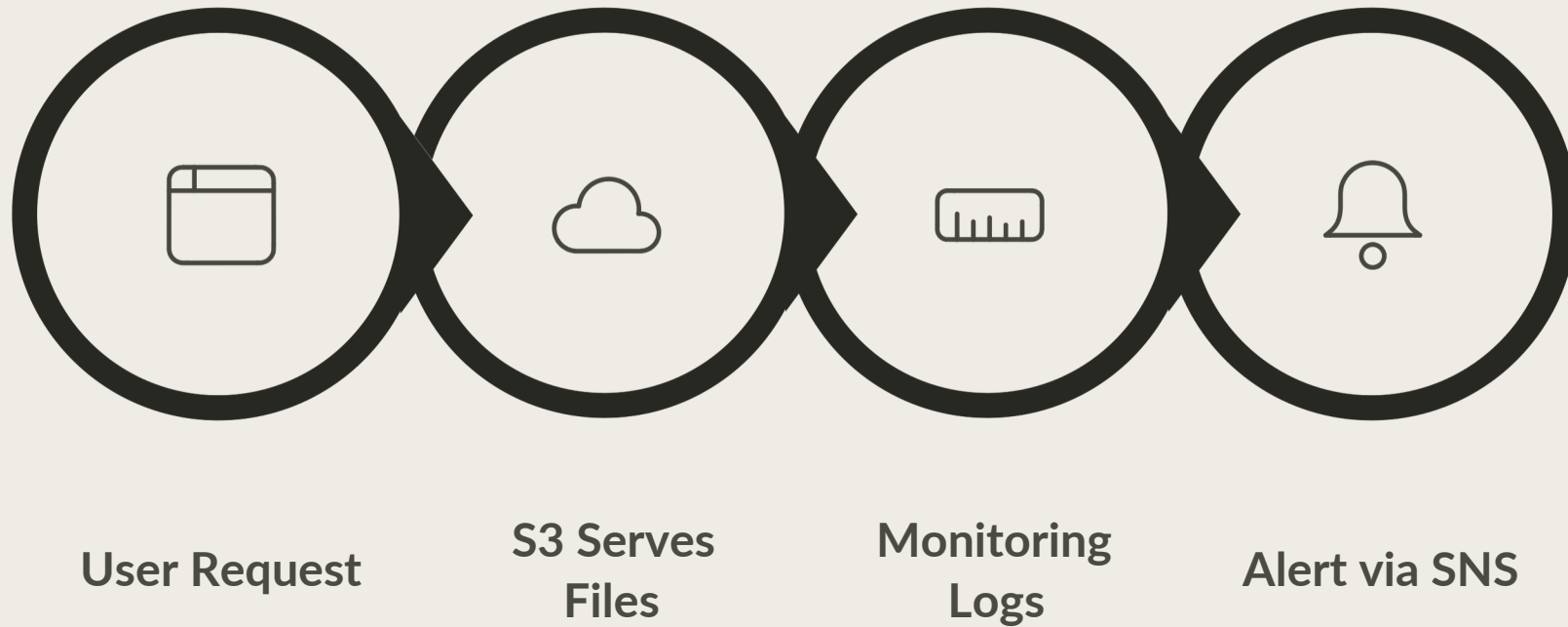


How It Works

Users access the portfolio website directly through the **S3 bucket endpoint**. CloudWatch monitors performance metrics, CloudTrail logs all API activity, SNS sends email alerts for critical events, and Systems Manager is planned for future infrastructure automation and management.

- ☐ All services are natively integrated within AWS, requiring no third-party tools or custom infrastructure.

Working of the System



The request-response cycle is fully serverless. S3 handles content delivery, while CloudWatch and CloudTrail operate silently in the background, ensuring every access and configuration change is monitored and logged without impacting performance.

Advantages & Applications



Key Advantages Real-World Applications

- Personal developer portfolios and resumes
- Product landing pages and marketing sites
- Documentation and project showcase pages
- Static frontends for serverless architectures

Highly Available & Durable

99.99% availability SLA with 11 nines of data durability.

Cost-Effective & Serverless

Pay only for storage and requests—no EC2 instances required.

Secure & Maintainable

Bucket policies, CloudTrail audit logs, and SNS alerts ensure security.

Challenges, Limitations & Future Scope

Current Limitations

Static Content Only

S3 serves only HTML, CSS, and JS — no server-side processing or databases.

CloudFront Unavailable

CDN deployment was restricted due to IAM permission constraints in the AWS Academy/Vocareum lab environment — a known limitation of the sandbox setup, not a design gap.

Future Enhancements

- **CloudFront CDN + Route 53**
Global content delivery with a custom domain.
- **SSL via AWS Certificate Manager**
HTTPS encryption for production readiness.
- **CI/CD with GitHub Actions**
Automated deployments on every code commit.
- **Lambda + DynamoDB + API Gateway**
Serverless backend for dynamic functionality.
- **Terraform Automation**
Infrastructure-as-Code for reproducible deployments.



Conclusion

This project demonstrates that a **production-ready static website** can be deployed on AWS with minimal cost and zero server management. By integrating S3, CloudWatch, CloudTrail, and SNS, the solution achieves **reliability, security, and observability** — a strong foundation for any cloud-native portfolio.

✓ Deployed

Static website live on Amazon S3

📊 Monitored

CloudWatch metrics active

🔒 Audited

CloudTrail logging enabled

🔔 Alerted

SNS notifications configured

THANK YOU

QUESTIONS?